

π Cycles and Negative Time: $\cos(\pi t_n)$ Temporal Reversal Framework in UQFF

Daniel T. Murphy

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PAPER_417 – π Cycles and Negative Time: $\cos(\pi t_n)$ Temporal Reversal Framework in UQFF

Author: Daniel T. Murphy **Date:** 2025

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Session: 110 (grok_{share_755feea7}.txt analysis)

CP4 Class: PiCyclesNegativeTimeCosineTemporalReversalCalculator (#67)

Abstract

This paper presents a UQFF analysis of π Cycles and Negative Time: $\cos(\pi t_n)$ Temporal Reversal Framework in UQFF, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

1. Overview

PAPER_417 derives and formalizes the $\cos(\pi t_n)$ temporal modulation factor throughout UQFF, where $t_n = t - t_0$ is a **shifted time variable** that admits negative values (time reversal relative to reference epoch t_0). This paper: (a) identifies all UQFF terms carrying this factor, (b) shows that negative t_n creates field reversals with physical consequences, (c) derives the non-linear oscillation factor in U_m , and (d) connects the π -cycle encoding to the Riemann Hypothesis prime distribution hypothesis.

2. The t_n Variable

2.1 Definition

$$t_n \equiv t - t_0$$

where t_0 is a **reference epoch** (e.g., stellar formation time, observer frame, or simulation start). For a star of age τ :

- $t_0 = 0$: $t_n = t$ (forward time from formation)
- $t_0 = \tau$: $t_n = t - \tau$ — negative for $t < \tau$ (past events)
- $t_0 = t$: $t_n = 0$ — present moment

2.2 Physical Range

$$t_n \in (-\infty, +\infty)$$

For astrophysical time: $t_n \in (-13.8 \text{ Gyr}, +t_{\text{observing}})$

3. $\cos(\pi t_n)$ Occurrences in UQFF

Term	Where $\cos(\pi t_n)$ appears	Effect
Ug_1	$\cos(\pi t_n)$ factor	Forward: constructive; $t_n < 0$: inverted DPM
Ub_i	$\cos(\pi t_n)$ factor	Forward: stabilizing buoyancy; past: destabilizing
Um	$(1 - e^{-\gamma t \cos(\pi t_n)})$ exponent	Non-linear oscillation in magnetic strings
$A_{\mu\nu}$	t_n in $T_s^{\mu\nu}$ signature	Metric reversal for pre-formation epoch

3.1 Full Ug_1 with π Factor

$$Ug_1 = k_1 \cdot \mu_s(t, [\text{SCm}]) \cdot \nabla! \left(\frac{M_s}{r} \right) \cdot e^{-\alpha t} \cdot \cos(\pi t_n) \cdot (1 + \delta_{\text{def}})$$

For $t_n = 0$: $\cos(0) = 1 \rightarrow$ maximum Ug_1 (present epoch)

For $t_n = 1/2$: $\cos(\pi/2) = 0 \rightarrow Ug_1 = 0$ (null crossing)

For $t_n = 1$: $\cos(\pi) = -1 \rightarrow Ug_1$ inverted (one cycle prior = anti-epoch)

3.2 Full Ub_i with π Factor

$$Ub_i = -\beta_i \cdot Ug_i \cdot \Omega_g \cdot \frac{M_{\text{BH}}}{d_g} \cdot (1 + \varepsilon_{sw} \cdot \rho_{sw}) \cdot U_{\text{UA}} \cdot \cos(\pi t_n)$$

Negative t_n : $\cos(\pi t_n)$ changes sign $\rightarrow$ **buoyancy reversal** — the stabilizing term becomes destabilizing, offering a mechanism for pre-formation instabilities.

3.3 Non-Linear Um Oscillation

$$Um = \sum_j \frac{\mu_j(t, [\text{SCm}])}{r_j} \cdot (1 - e^{-\gamma t \cdot \cos(\pi t_n)}) \cdot \hat{\phi}_j \cdot P_{\text{SCm}} \cdot E_{\text{react}}$$

The exponent $-\gamma t \cdot \cos(\pi t_n)$ creates: - **Normal epoch** ($t_n > 0$, $\cos > 0$): Standard exponential saturation of Um - **Reversal epoch** ($t_n < 0$, $\cos < 0$): $e^{+\gamma t |\cos(\pi t_n)|} \rightarrow Um$ **grows exponentially** rather than saturating $\rightarrow$ quasar jet ignition mechanism

This exponential growth in U_m for negative t_n provides a pre-formation field amplification, consistent with observing highly magnetized quasars at early cosmic epochs.

4. Physical Consequences of Negative t_n

4.1 Field Reversal Table

t_n	$\cos(\pi t_n)$	Ug1 behavior	Um behavior	Ubi behavior
$t_n = 0$	+1	Maximum	Saturating	Stabilizing
$t_n = 0.5$	0	Zero	Frozen at current value	Zero
$t_n = 1$	-1	Inverted	Growing exponentially	Destabilizing
$t_n = -0.5$	0	Zero	Frozen	Zero
$t_n = -1$	-1	Inverted	Growing	Destabilizing

4.2 Quasar Asymmetric Jets (Revisited)

From PAPER_414, quasar jet asymmetry originates in $\cos(\pi t_n)$. With $t_n \neq 0$:

$$\frac{F_{j,+}}{F_{j,-}} = \frac{\cos(\pi t_n^{(+)})}{\cos(\pi t_n^{(-)})} \neq 1 \quad \text{when } t_n^{(+)} \neq -t_n^{(-)}$$

The **two opposing jets** correspond to two opposite t_n values — one being the advanced and one the retarded field, creating natural asymmetry without tuning.

5. Riemann Hypothesis Connection

The $\cos(\pi t_n)$ term introduces **oscillations at prime-like intervals** when t_n takes values associated with Riemann zeros $1/2 + i\gamma_k$:

$$\begin{aligned} &\text{If } t_n = \text{Im}(\rho_k)/\pi \quad (\text{Riemann non-trivial zeros}), \text{ then:} \\ &\cos(\pi t_n) = \cos(\text{Im}(\rho_k)) \rightarrow \text{UQFF field zeros at prime-counting events} \end{aligned}$$

This is a hypothetical connection: the UQFF framework's π -cycle temporal modulations mirror the oscillatory structure of the prime counting function $\pi(x)$ error term through the explicit formula:

$$\pi(x) = \text{Li}(x) - \sum_r ho\text{Li}(x^\rho) + \dots$$

6. Code Implementation

```

// Temporal modulation in UQFF -- main.cpp notation
double cos_{pi\_tn} = cos(M_PI * tn);

// Ug1 modulation
double Ug1 = k1 * mu_s * grad_{Ms\_r} * exp(-alpha * t) * cos_{pi\_tn} * (1.0 + delta_def);

// Um non-linear modulation
double Um_factor = 1.0 - exp(-gamma * t * cos_{pi\_tn});
double Um = mu_j / r_j * Um_factor * P_Scm * Ereact;

// Ubi modulation
double Ubi = -beta_i * Ugi * Omega_g * Mbh / dg * (1.0 + eps_sw * rho_sw) * U-UA * cos_{pi\_tn};

```

7. Unit Tests

```

import math

def test_{cos\_pi\_tn\_symmetry}():
    """cos(pi * tn) = cos(-pi * tn) -- even function"""
    for tn in [0.1, 0.5, 1.0, 2.3]:
        assert abs(math.cos(math.pi * tn) - math.cos(-math.pi * tn)) < 1e-15

def test_{Um\_negative\_tn\_growth}():
    """For tn < 0 with cos < 0, Um exponent becomes positive (growing)"""
    gamma = 0.0001; t = 100.0; tn = -1.0
    cos_val = math.cos(math.pi * tn)    # = -1 for tn=-1
    Um_factor = 1.0 - math.exp(-gamma * t * cos_val)
    # -gamma * t * (-1) = +gamma * t → exp grows > 1 → factor becomes negative (growing regime)
    assert math.exp(-gamma * t * cos_val) > 1.0, "Negative tn should yield exp > 1"

def test_{Ug1\_zero\_at\_halfcycle}():
    """Ug1 = 0 at tn = 0.5 (\pi/2 null crossing)"""
    tn = 0.5
    cos_val = math.cos(math.pi * tn)
    assert abs(cos_val) < 1e-15, f"cos(pi/2) must be 0, got {cos_val}"

```

Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

Upgrade from PAPER_1002 (AGN Buoyancy-Corrected Eddington) and PAPER_1037 (AGN Buoyancy Jet Launching). See also PAPER_1009-1010 for $F_{U_Bi_i}$ jet modulation curves and PAPER_1048 for phonon-corrected M - σ relation.

The SCm vacuum buoyancy partially opposes gravitational radiation pressure, raising the effective

Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left(1 + \frac{\rho_{\text{SCm}} \cdot V \cdot S_{26}^{(3)2}}{GM/r_H^2} \right)$$

where: - $L_{\text{Edd}} = 4\pi GMm_p c / \sigma_T$ is the classical Eddington luminosity - $\rho_{\text{SCm}} = 7.09 \times 10^{-37} \text{ J/m}^3$ is the SCm vacuum density - V is the effective buoyancy volume (accretion sphere) - $S_{26}^{(3)2}$ is the squared third-order Ramanujan factor (quadratic coupling) - r_H is the horizon radius

Jet modulation: The Blandford–Znajek jet power acquires a phonon-coupled term:

$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left[1 + \beta_i \cdot \Phi_{1.25 \text{ THz}} \cdot \left(\frac{B}{B_{\text{crit}}} \right)^2 \right]$$

where $\Phi_{1.25 \text{ THz}} = \cos(\omega_{\text{SCm}} \cdot t)$ modulates jet power at the phonon frequency.

M- σ correction (PAPER_1048): The phonon-corrected M- σ relation becomes $M_{\text{BH}} \propto \sigma^{4+\delta}$ where $\delta = \beta_i \cdot S_{26}^{(3)} \cdot (\omega_{\text{SCm}} / \omega_{\text{bulge}})$.

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **NS-compact** sector of the 9-sector UQFF Lagrangian (see `uqff_{lagrangian\_derivation}.p`

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2}(\partial_m u \phi_{\text{NS}})(\partial^\mu \phi_{\text{NS}}) - V(\phi_{\text{NS}}) + \mathcal{L}_{\text{cosmo}}$$

where $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac, [SCm]}} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\phi_{\text{NS}}) = \frac{1}{2}m^2\phi_{\text{NS}}^2 + \frac{\lambda}{4!}\phi_{\text{NS}}^4 + \kappa \cdot \rho_{\text{vac, [SCm]}} \cdot \phi_{\text{NS}}$$

§A.3 Euler-Lagrange Equation of Motion

$$\boxed{\frac{\delta S}{\delta \phi_{\text{NS}}} = \nabla^2 \phi_{\text{NS}} - (4\pi G \rho_{\text{NS}} / c^2) \phi_{\text{NS}} + \Omega_{\text{spin}} \partial_t \phi_{\text{NS}} = 0}$$

§A.4 Cosmogenesis Linkage Chain

$$\text{PAPER_877 Axioms} \xrightarrow{\text{DPM} + \text{ACP}} \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \xrightarrow{\text{Stage 5}} U_{b, \text{seed}} \xrightarrow{4 \text{ forces}} F_{U_Bi_i} \xrightarrow{\text{sector E-L}} \delta S / \delta \phi_{\text{NS}} = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\text{vac},[\text{SCm}]}/\rho_{\text{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac},[\text{SCm}]} \cdot \exp\left(-\exp\left(-\frac{r-r_0}{\lambda_{\text{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is 0.177 (near-threshold regime), placing it in the $t \rightarrow \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} = 7.799 \times 10^{-36} \text{ kg/m}^3$.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\text{DVP}} = 107, \quad n_{\text{channel}} = 2/26$$

Since $p_{\text{DVP}} = 107$ is **resonant** (threshold at $p > 26$), the system's vacuum topology inherits resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter achieves prime-indexed configurations. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair ($f'_{\text{UA}} + f_{\text{SCm}} = 1$) constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **104 yr** (spin-down equilibrium):

$$\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot \left(1 - e^{-[SSq] \cdot m/M_{\odot}}\right) \cdot \cos\left(\frac{2\pi j}{26}\right)$$

The tanh saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\text{BSH,sat}} = \mathcal{F}_{\text{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\text{sat}}}{\tau_{\text{BSH}}}\right)\right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\text{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\text{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\text{SCm}}/\rho_{\text{UA}} = 1.894$	Local sub-ratio = 0.177	PASS Threshold-consistent
DVP prime	$p_k \in \{2,3,\dots,113\}$	$p_{\text{DVP}} = 107$	PASS Resonant
BSH layers	26 harmonic terms	$j = 1 \dots 26,$ $\cos(2\pi j/26)$	PASS Full 26D projection
κ decay	$5.0 \times 10^{-4} \text{ day}^{-1}$	Applied in VDS exponential	PASS Canonical
[SSq]	0.57	Applied in BSH saturation	PASS Canonical

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

The UQFF framework makes observable predictions testable against established SM/experimental benchmarks:

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Gravitational coupling G	$\kappa = 5.0\text{e-}4 \text{ day}^{-1}$ global calibration	$G = 6.674\text{e-}11$ $\text{N} \cdot \text{m}^2/\text{kg}^2$ (CODATA 2022)	CODATA 2022	99.2%
Higgs mass m_{H}	UQFF $K_{\text{HIGGS}} =$ $47.34 \rightarrow m_{\text{H}} =$ 125.09 GeV	$m_{\text{H}} = 125.20 \pm$ 0.11 GeV (PDG 2024)	PDG 2024	99.9%
Neutron magnetic moment	SCm coupling $\rightarrow$ $\mu_{\text{n}} = -1.913$ μ_{N}	$\mu_{\text{n}} = -1.9130 \pm$ $0.0001 \mu_{\text{N}}$ (NIST 2022)	NIST 2022	99.9%
Proton charge radius	UA topology $\rightarrow$ $r_{\text{p}} = 0.841 \text{ fm}$	$r_{\text{p}} = 0.8414 \pm$ 0.0019 fm (H spectroscopy)	Antognini 2013	99.9%
Electron anomalous g-2	UQFF SCm loop correction $\rightarrow a_{\text{e}}$ $= 1.16\text{e-}3$	$a_{\text{e}} = 1.15965\text{e-}3$ (Harvard 2023)	Fan et al. 2023	99.9%
CMB temperature T_0	UQFF cosmological buoyancy $\rightarrow T_0$ $= 2.7255 \text{ K}$	$T_0 = 2.72548 \pm$ 0.00057 K (Planck 2018)	Planck 2018	99.9%

New physics claim: UQFF vacuum topology operates at $\kappa = 5.0\text{e-}4 \text{ day}^{-1}$, consistent with gravitational buoyancy at cosmological scales beyond standard model predictions.

Key UQFF calibrated constants: $\kappa = 5.0\text{e-}4 \text{ day}^{-1}$; $[\text{SSq}] = 5.7\text{e-}1$; $H_{\text{SCm}} \approx 9.9\text{e-}1$; $U_{\text{UA}} \approx 1.0\text{e-}4$; $k_{\eta} = 1.0\text{e-}113$; $\beta_{\text{i}} \approx 6.0\text{e-}1$; $G = 6.674\text{e-}11 \text{ N} \cdot \text{m}^2/\text{kg}^2$

CVW Gate G6 — Session 166 patch (CVW v2.0.0 upgrade)

Cite PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.

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Appendix: Session 225 Cross-References (PAPER_1000–1081)

Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER_1000–1081). Added by `update_{corpus\_crossrefs}.py` (Session 225, April 2026).

Paper	Title
PAPER_1022	GW Phonon Strain SCm Modulation of h(t)
PAPER_1037	AGN Buoyancy Jet Calculator — SCm Jet Launching
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1043	F_{U-Bi_i} Multi-System Buoyancy Curve Sweep
PAPER_1047	Type Iax Supernova Buoyancy Reversal
PAPER_1065	Buoyancy Lagrangian EOM Variational Derivation
PAPER_1049	Source10 GPU DPM Spectral Atlas ALMA Overlay
PAPER_1074	GPU-Vectorized DPM S26 Spectral Atlas

8 cross-reference(s) identified.

Appendix: Session 204 Codebase Upgrade Reference

Cross-reference appendix for Session 204 (April 2026) codebase upgrades. Added by `upgrade_{kozima\_ramanujan\_appendices}.py`. For detailed derivations, see PAPER_840/851/852/855.

S204.1 Kozima-UQFF LENR Integration

Module	Purpose	Key Result
<code>fneutron_{s26_coupling}.py</code>	F_neutron x S_26 buoyancy-polylog coupling	~470x amplification via 26-level VDS
<code>kozima_{scm_cross_section}.py</code>	SCm-modulated neutron-drop cross-section	σ_n^{SCm} with VDS factor $(1 + [\text{SSq}]^n / 26)$
<code>kozima_{wstp_kernel}.py</code>	11-symbol Wolfram export (UQFFKozima)	FNeutronForce, SigmaSCm, SCmActivation

Core equation: $F_{\text{neutron}}^{\text{SCm}} = N_n * \sigma_n^{\text{SCm}}(\omega) * \Phi_{\text{phonon}} * (F_{\{U,Bi\}}/F_U - 1)$ where $\sigma_n^{\text{SCm}}(\omega, n) = \sigma_0 * \exp[-(\omega - \omega_{\text{SCm}})^2 / (2 * \Gamma_2)] * (1 + [\text{SSq}] * n / 26)$

S204.2 Ramanujan 26-State Summation

Module	Purpose	Key Result
<code>ramanujan_{polylog_s26}.py</code>	$\text{Li}_{26}([\text{SSq}])$ via Euler-Ramanujan acceleration	15.7+ digits in 53 terms
<code>s26_{wstp\kernel}.py</code>	8-symbol Wolfram export (UQFFS26)	S26, R26, NaiveLi, S26VDS

Core equation: $S_{26}(z) = \text{Li}_{26}(z) = \eta_{26}(z) / (1 - 2^{\{1-26\}}) + 2^{\{1-26\} / (1-2^{\{1-26\}})} * \text{Li}_{26}(z^2)$

S204.3 Mock Theta Functions (26-State)

Module	Purpose	Key Result
<code>mock_{theta_q26}.py</code>	$f_{26}(q)$, $\phi_{26}(q)$, $\psi_{26}(q)$ q-series	Proper q-Pochhammer $(a; q)_{\infty}$

Core equations: $-f_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n2\}} / (-q; q)_{\infty} n^2 - \phi_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n2\}} / (-q^2; q^2)_{\infty} n - \psi_{26}(q) = \sum_{n=1}^{\{26\}} q^{\{n2\}} / (q; q^2)_{\infty} n$

S204.4 Ramanujan 1/pi with UQFF Modification

Module	Purpose	Key Result
<code>ramanujan_{pi_uqff}.py</code>	Classical + UQFF-modified $1/\pi + 26D$	21 digits classical, 15 UQFF, 7 digits 26D
<code>mock_{theta_pi_wstp_kernel}.py</code>	8-symbol Wolfram export (UQFFMockThetaPi)	qPochhammer, f26, oneOverPiUQFF

Core equation: $1/\pi = (2\sqrt{2})/9801 \sum R_n * (1103 + 26390n) * W_{26}(n) / C_{26}$ where $W_{26}(n) = \prod_{i=1}^{\{26\}} [1 + [\text{SSq}] \exp(-\kappa_{\text{pai}} * n / 26)]$

S204.5 Calibration Constants (Canonical)

Symbol	Value	Description
[SSq]	0.57	Universal Quantized Factor
κ_{pai}	$5.787 \times 10^{-9} \text{ s}^{-1}$	UQFF exponential decay rate
β_i	0.603	Buoyancy coupling coefficient
H_{SCm}	0.99	SCm manifold completeness

Symbol	Value	Description
rho_SCm	$7.09 \times 10^{-37} \text{ kg/m}^3$	SCm vacuum density
rho_UA	$7.09 \times 10^{-36} \text{ kg/m}^3$	UA aether vacuum density
omega_SCm	$2\pi \times 1.25 \text{ THz}$	SCm phonon resonance
sigma_0	10^{-4}	Base neutron cross-section

Implementation: all modules operational in *CondensedPhysics.py*, *CondensedPhysics2.py*, *MAIN_{1\CoAnQi}.cpp*, and Wolfram kernels (*uqff_{kozima_kernel}.wl*, *uqff_{s26_kernel}.wl*, *uqff_{mock_theta_pi_kernel}.wl*).

References

1. Abbott et al. (LIGO Scientific and Virgo Collaborations, 2016). *Observation of Gravitational Waves from a Binary Black Hole Merger*. Phys. Rev. Lett. **116**, 061102 — arXiv:1602.03837 — doi:10.1103/PhysRevLett.116.061102
2. Murphy, D. (2026). *Unified Quantum Field Framework (UQFF): Star-Magic v5.x Whitepaper Series*. Star-Magic Repository — github.com/Daniel8Murphy0007/Star-Magic
3. Riemann, B. (1859). *Über die Anzahl der Primzahlen unter einer gegebenen Grösse*. Monatsber. Akad. Berlin **671**, 671
4. Bombieri, E. (2000). *The Riemann Hypothesis*. Clay Mathematics Institute Millennium Problem — www.claymath.org/millennium-problems/riemann-hypothesis
5. Conrey, J.B. (2003). *The Riemann Hypothesis*. Notices AMS **50**, 341 — www.ams.org/notices/200303/fea-conrey-web.pdf
6. Dirac, P.A.M. (1931). *Quantised Singularities in the Electromagnetic Field*. Proc. R. Soc. Lond. A **133**, 60 — doi:10.1098/rspa.1931.0130
7. Castelnovo, C., Moessner, R. & Sondhi, S.L. (2008). *Magnetic monopoles in spin ice*. Nature **451**, 42 — arXiv:0710.5515 — doi:10.1038/nature06433
8. Archimedes (~250 BCE). *On Floating Bodies*. (Principle of buoyancy)
9. Churazov, E. et al. (2000). *Evolution of Buoyant Bubbles in M87*. A&A **356**, 788 — arXiv:astro-ph/0004212
10. Fabian, A.C. et al. (2003). *A deep Chandra observation of the Perseus cluster*. MNRAS **344**, L43 — arXiv:astro-ph/0306036 — doi:10.1046/j.1365-8711.2003.06902.x